

Numerical Modeling of Granular Physics

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1 Introduction

The purpose of this numerical model is to recreate a lab experiment landslide model. The lab model's purpose is to calculate the stress along the bottom boundary of a landslide. The lab landslide is modeled with a box that is filled with grains, which is then slowly tilted until a landslide occurs. In order to model this, we use a numerical model to describe the mechanical behavior of assemblies of spheres or discs.[1] The lab model is restricted to calculating displacements and stress at the boundaries of the landslide model. The numerical model provides a way for us to get displacements and stresses from the interior of the landslide model.

2 Granular Physics

In this numerical model, we simulate granular materials by treating individual particles as circular grains. Each grain is represented by its physical properties such as position, radius, mass, velocity, orientation, and angular velocity. Contact forces between grains are calculated when overlaps occur due to collisions or interactions with boundaries. Using equations of linear and angular dynamics, the forces and torques are computed and applied to each particle at each time step.

2.1 Linear Dynamics

The linear equations of motion are derived from Newton's Second Law,

$$\vec{F}_{\text{ext}} = m\vec{a} = m\frac{d\vec{v}}{dt} \quad (1)$$

where $\vec{F}_{\text{ext}}$ is the total external force acting on the particle, m is the mass of the particle, $\vec{a} = \frac{d\vec{v}}{dt}$ is the particle's acceleration, $\vec{v} = \frac{d\vec{x}}{dt}$ is the velocity.

The linear state update for position and velocity follows from Euler integration:

$$\begin{aligned} \vec{v}_{\text{new}} &= \vec{v}_{\text{old}} + \frac{\vec{F}_{\text{ext}}}{m}\Delta t \\ \vec{x}_{\text{new}} &= \vec{x}_{\text{old}} + \vec{v}_{\text{new}}\Delta t \end{aligned} \quad (2)$$

2.2 Angular Dynamics

The rotational dynamics follow from the angular equivalent of Newton's Second Law:

$$\tau_{\text{ext}} = I\alpha = I \frac{d\omega}{dt} \quad (3)$$

where τ_{ext} is the external torque acting on the particle, $I = \frac{1}{2}mr^2$ is the moment of inertia of a solid disk, $\alpha = \frac{d\omega}{dt}$ is the angular acceleration, ω is the angular velocity.

Torque is calculated based on the applied tangential force:

$$\tau_{\text{ext}} = \vec{r} \times \vec{F}_t \quad (4)$$

where $\vec{r}$ is the vector from the particle's center to the contact point, $\vec{F}_t$ is the tangential force. The cross product represents the magnitude of the torque.

The angular state update follows:

$$\begin{aligned} \omega_{\text{new}} &= \omega_{\text{old}} + \frac{\tau_{\text{ext}}}{I} \Delta t \\ \theta_{\text{new}} &= \theta_{\text{old}} + \omega_{\text{new}} \Delta t \end{aligned} \quad (5)$$

This approach ensures realistic particle interactions while maintaining numerical stability.

3 Methods

3.1 First Order Euler Time Integration

To integrate these equations into the numerical model, we use a first-order Euler Time Integration, where we take a small time step and calculate the next value by adding the derivative to its current value. This allows us to update the dynamics of each particle. We choose each particle's (P) initial condition (P_{old}) and update the conditions each time step (Δt) for the chosen duration (T). While the simulation time (t) is less than T :

We update the particles conditions:

$$P_{\text{new}} = P_{\text{old}} + H(P_{\text{old}}, t \cdot \Delta t) \quad (6)$$

advance the time:

$$t = t + \Delta t \quad (7)$$

We then return the updated: t, P_{new} and update the conditions: $P_{\text{old}} = P_{\text{new}}$.

3.2 Numerical Model

We implemented the numerical model using C, leveraging the First-Order Euler Time Integration method to simulate particle interactions, dynamics, and boundary conditions. We gave each particle initial conditions: position (x, y), radius (r), angular velocity (ω), orientation angle (θ), mass (m), velocities (v_x, v_y), and moment of inertia for a disc ($I = \frac{1}{2}mr^2$).

We update each particle's state using the Euler Time integration as demonstrated in equation: (6). where:

$$H(\mathbf{P}) = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{v}_x \\ \dot{v}_y \\ \dot{\theta} \\ \dot{\omega} \end{bmatrix} \quad (8)$$

is the state update function calculated from force interactions, boundary conditions, and contact forces. We output each of the particle's conditions for every time step to prepare for visualization.

3.3 Visualization

To verify the results of our numerical model, it was visualized using Python. Loading the output from the C simulation, we read the particles conditions. We plotted the x and y positions on a graph and gave it a radius r, making a circle. To add the orientation angle, we plotted a line that goes from (x, y) to $(x + r \cos(\theta), y + r \sin(\theta))$. This line dynamically updated with each time step, illustrating the particles' rotational motion. Frames were generated for each time step and saved as high-resolution images, then converted to and animated as a GIF. This visualization framework enabled both qualitative and quantitative analysis of particle dynamics, helping to identify potential areas for numerical improvement in the C simulation.

4 Results

4.1 Satellite Orbit Model

We begin by modeling a satellite orbiting the Earth at an altitude of 1000 kilometers. The satellite is initialized with $x = 0$ km, $y = 1000$ km, $v_x = 0$, $v_y = 0$, $m = 1.5$ kg, mass of the Earth $M = 5.97219 \times 10^{24}$ kg.

The gravitational force acting on the satellite is given by:

$$F = \frac{GmM}{r^2}, \quad (9)$$

where: $G = 6.67430 \times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2}$ is the gravitational constant, m is the mass of the satellite, M is the mass of the Earth, $r = \sqrt{x^2 + y^2}$ is the distance between the satellite and the Earth (assumed to be at the origin).

Using this equation, we update the satellite's position and velocity at each time step. Specifically:

$$a_x = -\frac{GMx}{r^3}, \quad a_y = -\frac{GM y}{r^3}, \quad (10)$$

where a_x and a_y are the accelerations in the x and y directions, respectively.

The velocities and the positions are updated as seen in Eqn (2).

For this scenario, we expect the satellite to orbit the Earth (represented by the origin) in a nearly perfect circle. The results of the simulation, as visualized in Figure 1, confirm this expectation.

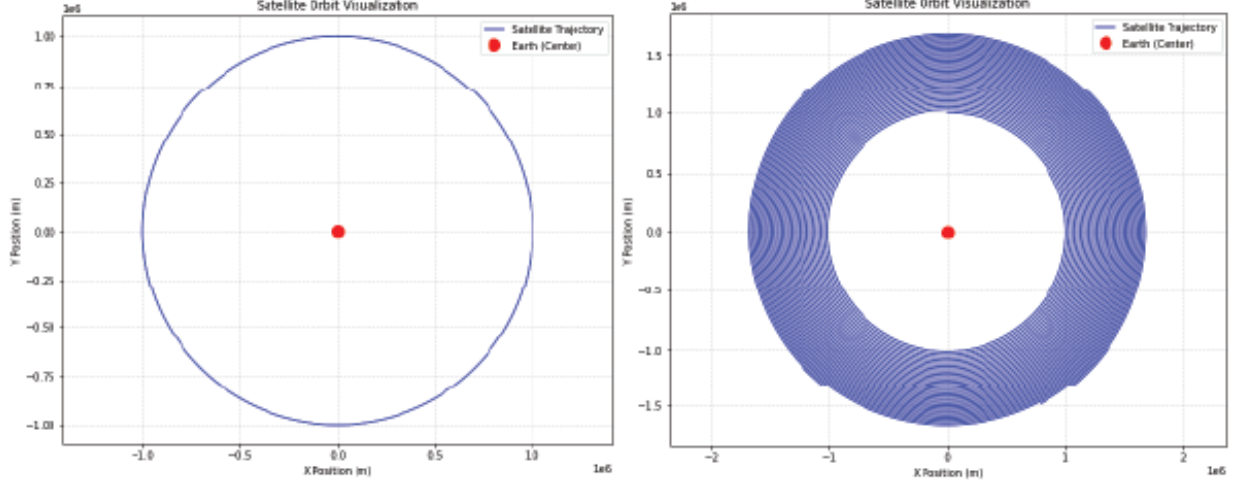


Fig. 1: Visualization of the satellite's position over time, where the blue line represents the satellite's path and the red line the earth. The figure on the right represents $\Delta t * 10$ demonstrating that a smaller the time step will generate a more accurate model.

4.2 Ball bouncing and spinning

To model a bouncing ball we initialize the conditions to be:

$$P_{\text{initial}} = \begin{bmatrix} r \\ m \\ x \\ y \\ v_x \\ v_y \\ \theta \\ \omega \end{bmatrix} = \begin{bmatrix} 0.01 \\ 0.2 \\ 0 \\ 0.2 \\ 1 \\ -2 \\ \frac{\pi}{4} \\ 0 \end{bmatrix} \quad (11)$$

with the force of gravity acting on it: The external force is represented by:

$$F_{\text{ext}} = \begin{bmatrix} 0 \\ -mg \end{bmatrix} \quad (12)$$

In this model, we model contact between the particle and the ground. We detect contact between the particle and the ground at $y = 0$ using:

$$y - r < 0 \quad (13)$$

for each time step.

The normal force is given by:

$$F_n = k_n \delta_n - c_n v_n \quad (14)$$

where k_n is the normal stiffness constant, c_n is the normal damping constant, and v_n is the velocity component in the normal direction.

Similarly, the tangential force is calculated as:

$$F_t = k_t \delta_t + c_t v_t \quad (15)$$

where k_t and c_t are the tangential stiffness and damping constants, respectively, and v_t is the tangential velocity.

The tangential displacement δ_t accumulates over time as the ball slides along the ground.

$$\delta_t = v_t \Delta t \quad (16)$$

To account for friction, a threshold is applied to the tangential force:

$$|\mathbf{F}_t| \leq \mu_s |\mathbf{F}_n| \quad (17)$$

If the tangential force exceeds this threshold, it is scaled back:

$$\mathbf{F}_t = \mu_s |\mathbf{F}_n| \frac{\mathbf{F}_t}{|\mathbf{F}_t|} \quad (18)$$

Torque is then calculated using:

$$\tau_{\text{ext}} = \vec{r} \times \vec{F}_t \quad (19)$$

where $\vec{r}$ is the vector from the ball's center to the point of contact.

The total external forces and torques acting on the ball are then used to update its linear and angular velocities.

$$\mathbf{v} = \frac{\mathbf{F}_{\text{ext}}}{m} \quad \text{and} \quad \omega = \frac{\tau_{\text{ext}}}{I} \quad (20)$$

The ball's position and orientation are updated for each time step using first-order Euler integration as seen in equations(2) and (5). This model captures both the linear and rotational dynamics of the bouncing ball. When initialized with a velocity directed downward and to the right, the ball is expected to bounce and gradually come to rest, as demonstrated in Figure 2.

4.3 Particle-Particle Interactions

In our simulation, particle-particle interactions are calculated using a linear spring-dashpot model. The interaction forces are computed whenever two particles overlap. We define the overlap δ_n between two particles i and j as:

$$\delta_n = (r_i + r_j) - |\vec{x}_i - \vec{x}_j| \quad (21)$$

where r_i, r_j are the radii of particles of i and j and $(\vec{x}_i, \vec{x}_j)$ are the positions of particles i and j .

If $\delta_n > 0$, a contact force is applied between the particles. The normal force $\vec{F}_n$ is computed based on the overlap using a spring-dashpot model from Eq. (14). The unit normal vector $\hat{n}$ points from particle i to particle j :

$$\hat{n} = \frac{\vec{x}_j - \vec{x}_i}{|\vec{x}_j - \vec{x}_i|} \quad (22)$$

The relative velocity between the two particles is:

$$\vec{v}_{\text{rel}} = \vec{v}_j - \vec{v}_i + \vec{r}_i \times \omega_i - \vec{r}_j \times \omega_j \quad (23)$$

The velocity component along the normal direction is:

$$v_n = \vec{v}_{\text{rel}} \cdot \hat{n} \quad (24)$$

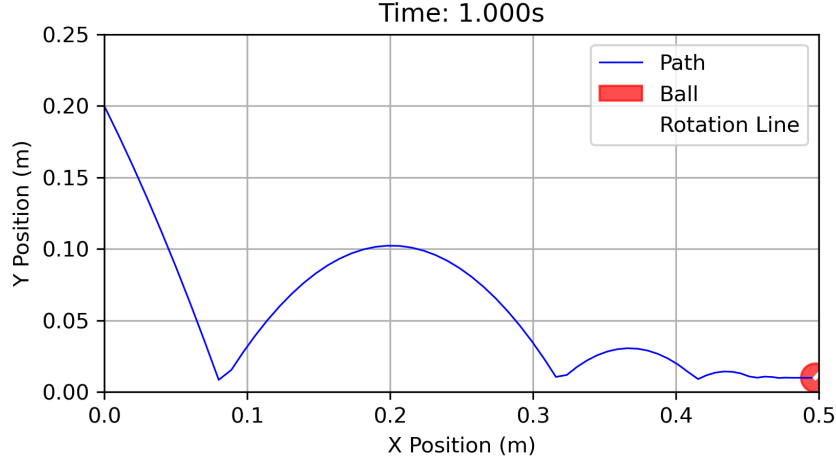


Fig. 2: Shows the path of the bouncing ball as it bounces and comes to rest.

Thus, the normal force becomes:

$$\vec{F}_n = -k_n \delta_n \hat{n} - c_n v_n \hat{n} \quad (25)$$

To visualize the particle-particle contact, we drop particles on top of each other; we expect the particles to bounce then settle on top of each other, which we see in Figure 3.

5 Conclusion

This numerical model provides a framework for simulating the dynamics of particles in a granular system, including both linear and rotational motions. While the current implementation successfully handles particle-wall interactions and computes forces such as the normal and tangential components, some limitations remain. Future work will focus on enhancing the model in the many ways. Incorporate tangential forces into particle-particle interactions to capture frictional effects more accurately. This enhancement will allow the simulation to model realistic scenarios such as shear deformation and energy dissipation during collisions. Address known issues, such as the floor spinning behavior (Figure 4), to ensure the stability and accuracy of the simulation. Debugging key components, like torque calculations and contact point adjustments, will be prioritized to resolve these anomalies. Optimization for Large Systems Improve computational efficiency

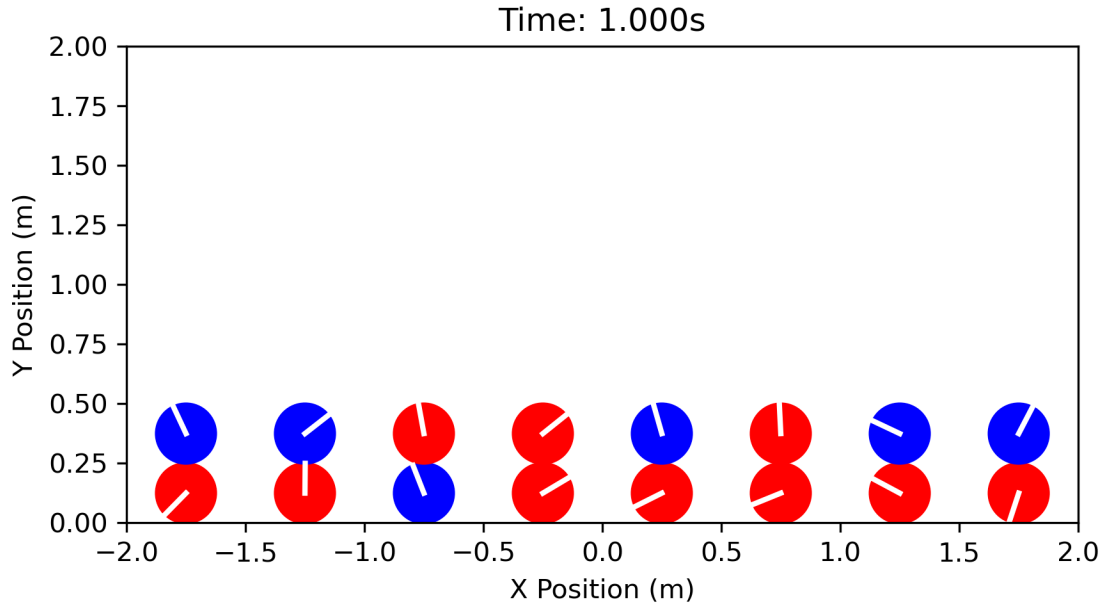


Fig. 3: Visualization of particles resting on top of each other showing the particle-particle contact interaction.

to handle larger particle assemblies. Optimizations could include spatial partitioning methods (e.g., grid-based collision detection) to reduce the computational cost of interaction calculations. Compare simulation results with experimental data and theoretical models to validate the accuracy of the forces, torques, and resulting dynamics. By addressing these areas, the model can be refined into a robust tool for studying granular materials and building the landslide model.

References

- [1] Stefan Luding. “Introduction to Discrete Element Methods: Basic of Contact Force Models and How to Perform the Micro–Macro Transition to Continuum Theory”. In: *European Journal of Environmental and Civil Engineering* 12.7–8 (2008), pp. 785–826. DOI: 10.1080/19648189.2008.9693050.

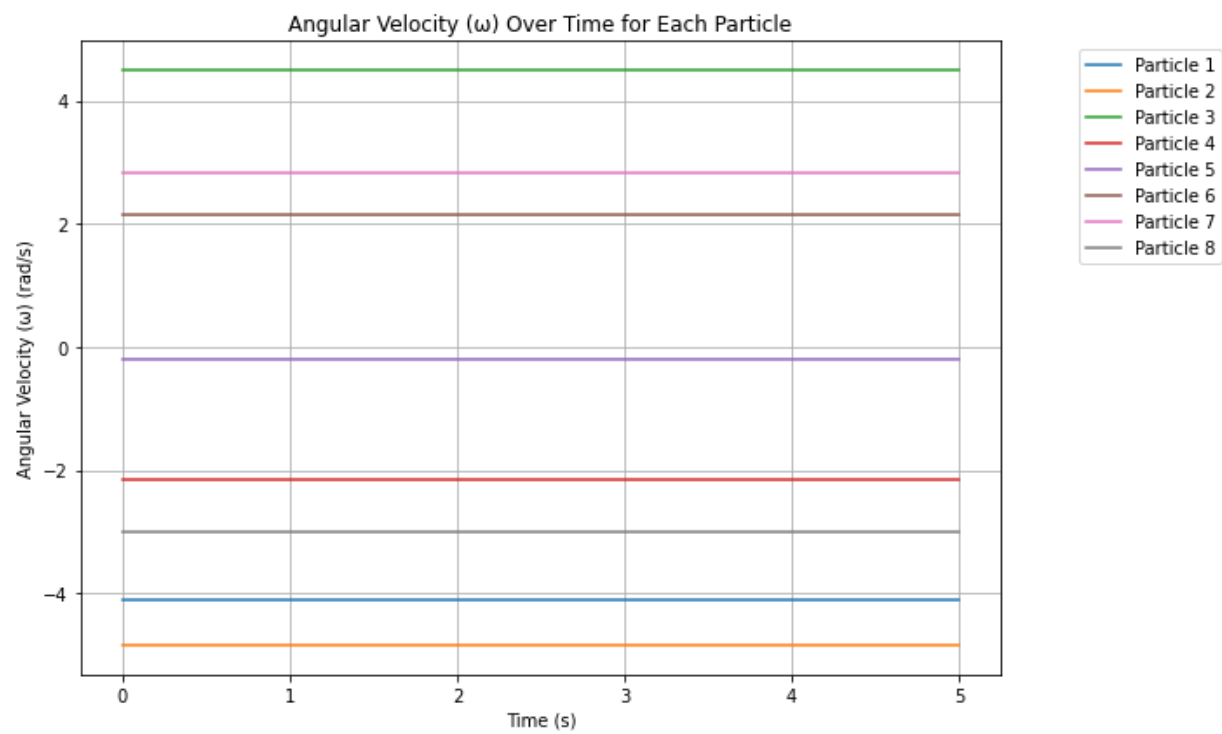


Fig. 4: This figure shows the angular velocity of the particles on the ground remaining nonzero and constant instead of slowing down towards zero as we would expect.

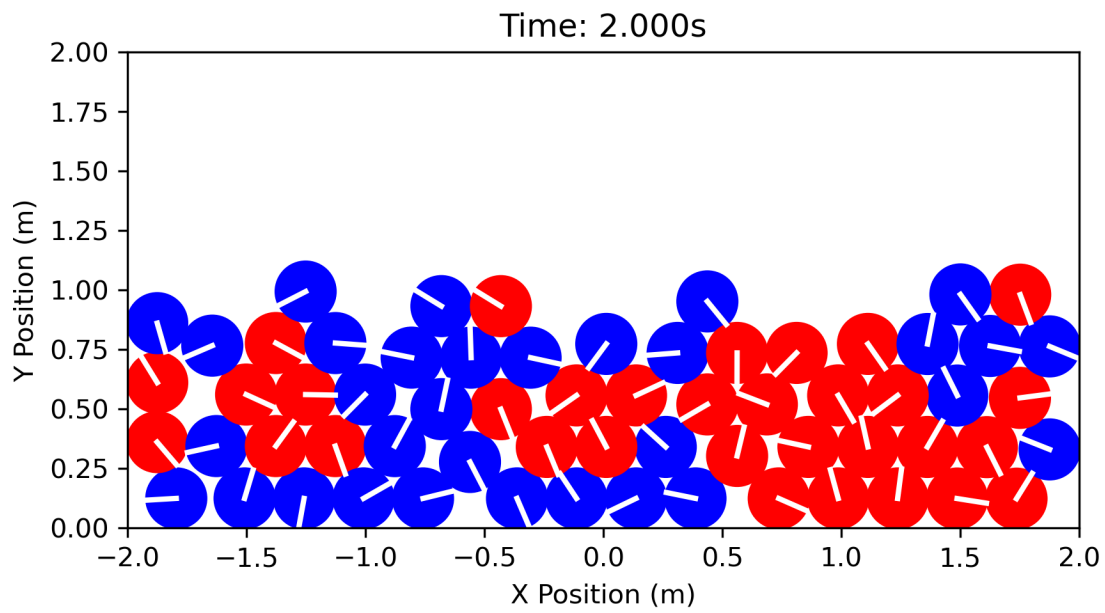


Fig. 5: Visualization of 64 grains settling into a simulated box, representing the process of grains being poured into the landslide experiment setup